

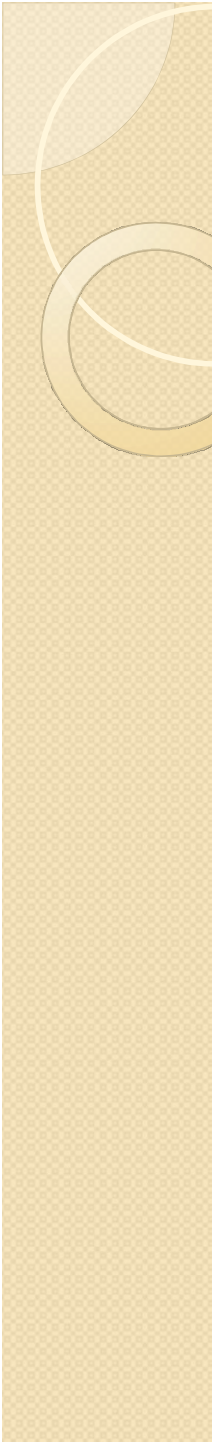


Lecture 7:-

Natural Language Processing (NLP)

How Computers Understand Human Language

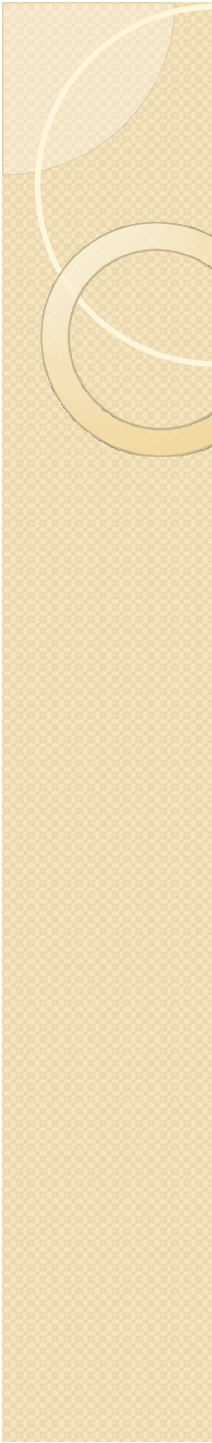
<https://www.lexalytics.com/>



Learning Outcomes

Students will be able to:

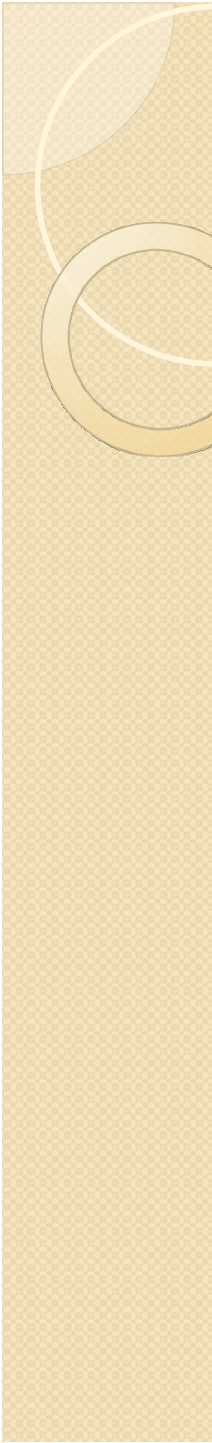
- Explain NLP
- Understand Tokenization
- Understand Embeddings
- Perform Sentiment Analysis
- Perform Named Entity Recognition (NER)
- Translate text using AI
- Explore Hugging Face Models and Spaces
- Relate every concept to ChatGPT



Activity:- How does Google Translate understand all these languages?

Good morning

How many languages can you say this in?

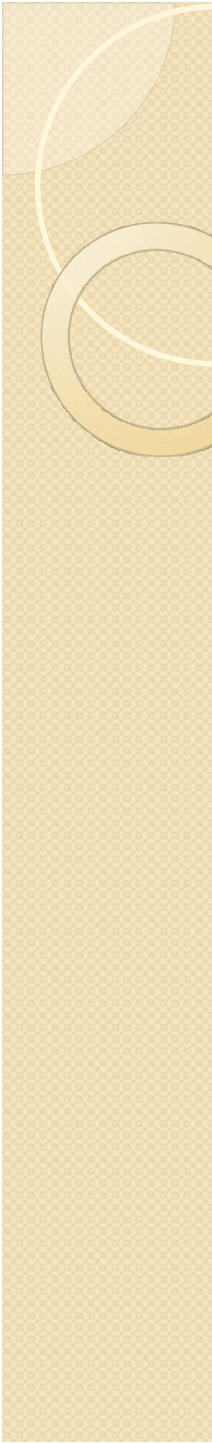


What is NLP?

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, generate and respond using human language.

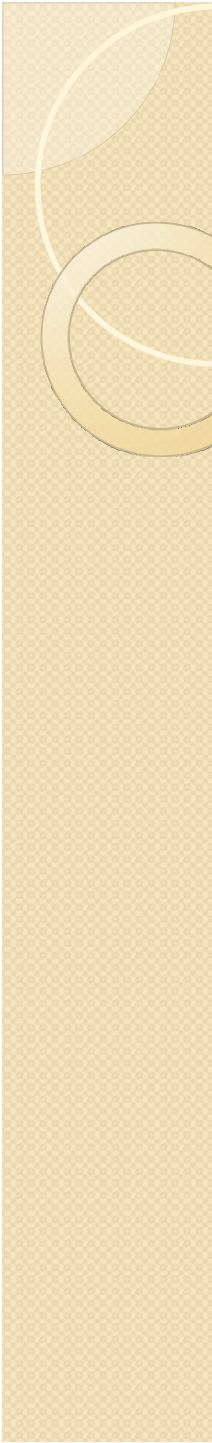
Examples

- ChatGPT
- Gemini
- Google Translate
- Siri
- Alexa
- Grammarly
- Gmail Spam Filter



Activity

One app they use daily that understands language.



How NLP Works

Human Language



Cleaning



Tokenization



Embeddings



Transformer Model

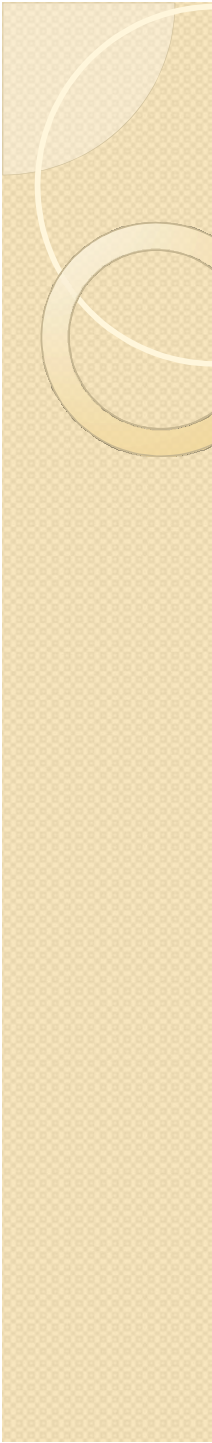


Prediction



Response

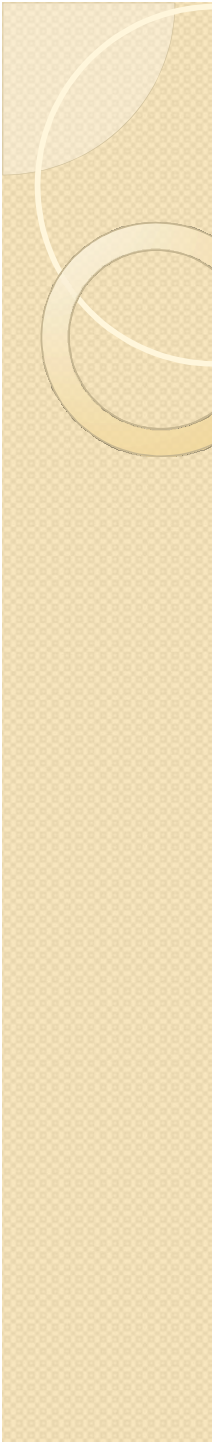
"This is exactly what happens inside ChatGPT."



Activity:-

Artificial Intelligence is changing education rapidly.

- Break It into tokens



Artificial
Intelligence
is
changing
education
rapidly
.

What is Tokenization?

Definition

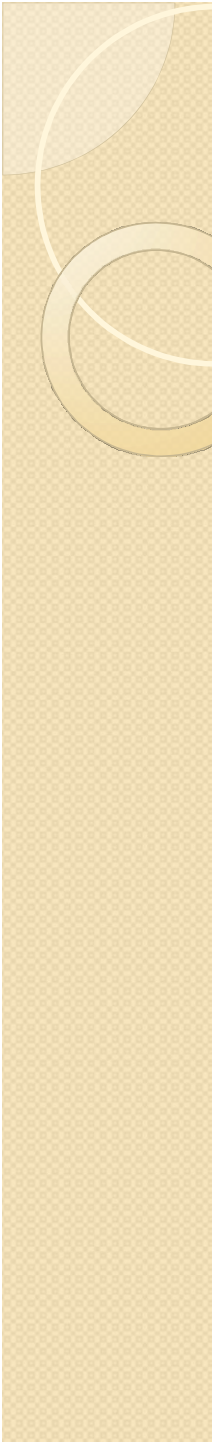
Tokenization is the process of breaking text into smaller pieces called **tokens**.

Example

I use AI.

↓

[I , use, AI]



Live Demo

Open the **OpenAI Tokenizer**
(or another tokenizer demo if available).

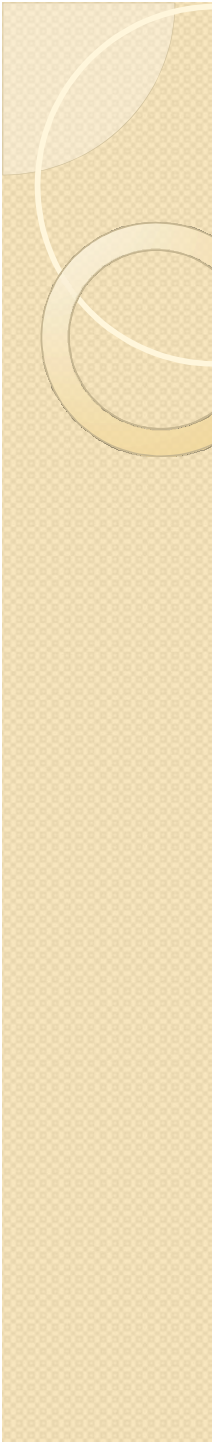
Paste different prompts.

Compare:

Short prompt

Long prompt

Q- why token count matters.



Activity: Prompt Competition

Prompt:

Write a Java Program to Reverse Linked List.

Rewrites the prompt to make it more specific.

Then compare:

- * Which prompt is longer?
- * Which uses more tokens?

How prompt quality affects output.

What are Embeddings?

Definition

Embeddings convert tokens into numerical vectors that capture their meaning.

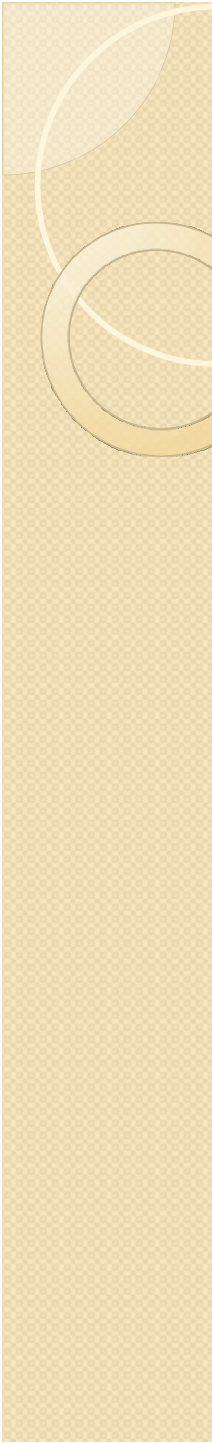
Example:

Cat

Dog

Tiger

These words are close in meaning, so their embeddings are also close.



Activity: Human Embeddings

Following word cards:

Dog

Cat

Lion

Car

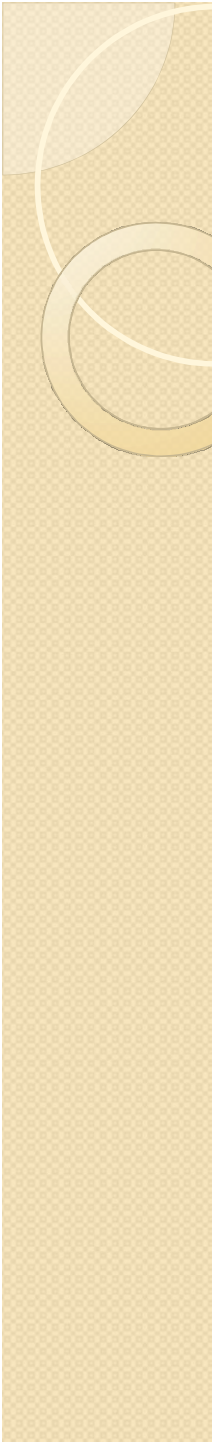
Bus

Bike

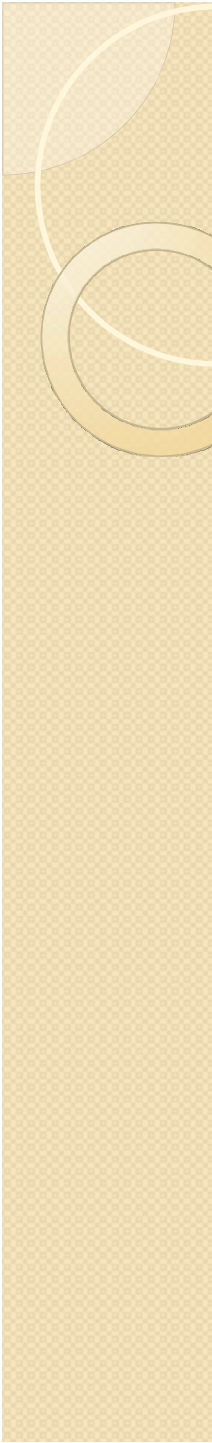
Apple

Mango

Make groups of similar meaning.



"Your brain grouped them based on meaning. AI does the same mathematically using embeddings."



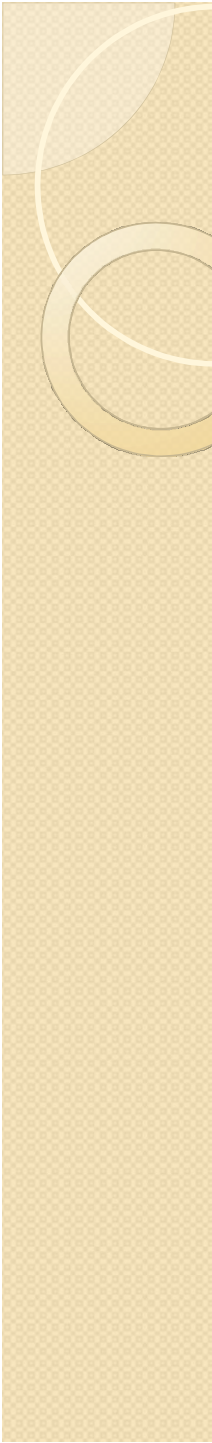
Visualizing Embeddings

Show a 2D embedding diagram with clusters.

why:

- * Cat is closer to Dog than to Airplane.
- * King and Queen are closer than King and Banana.

<https://www.adityabawankule.io/tools/embedding-playground>



Sentiment Analysis

Definition

Sentiment Analysis determines whether text expresses a **positive**, **negative**, or **neutral** opinion.

Examples:

"This course is amazing." → Positive

"The internet is very slow today." → Negative

"Today is Monday." → Neutral

Activity: Judge the Emotion

Display 10 sentences.

Each student raises:

- 👍 Positive
- 👎 Negative
- 👉 Neutral

Example sentences:

I love ChatGPT.

The food was terrible.

This phone is okay.

The class is interesting.

I am confused.

<https://huggingface.co/spaces/nazianafis/Sentiment-Analysis>

Named Entity Recognition (NER)

Definition

NER identifies important entities in text, such as:

- Person
- Organization
- Location
- Date
- Money

Example:

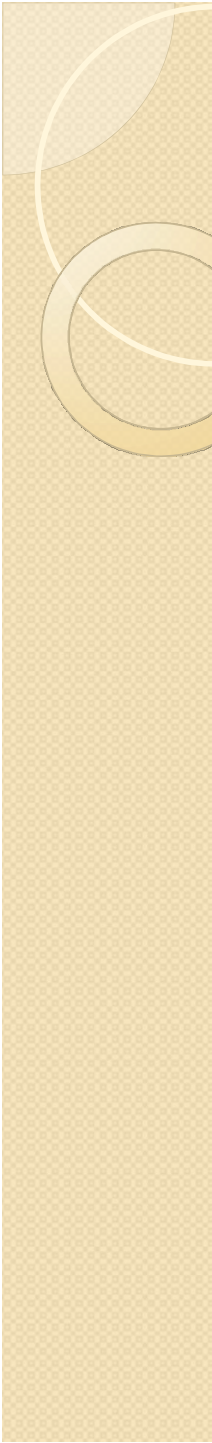
Faraz Alam teaches at Buddha Institute
of Management in Gorakhpur.

Entities:

Faraz Alam → Person

Buddha Institute of Management → Organization

Gorakhpur → Location



Slide 14 Activity: Become the AI

Give some sentences, find the entities in following:-

Then compare with a Hugging Face NER demo.

https://huggingface.co/spaces/koakande/named_entity_recognition

Slide 15 Machine Translation

Definition

Machine Translation automatically converts text from one language to another.

Examples:

English → Hindi

Hindi → Urdu

English → French

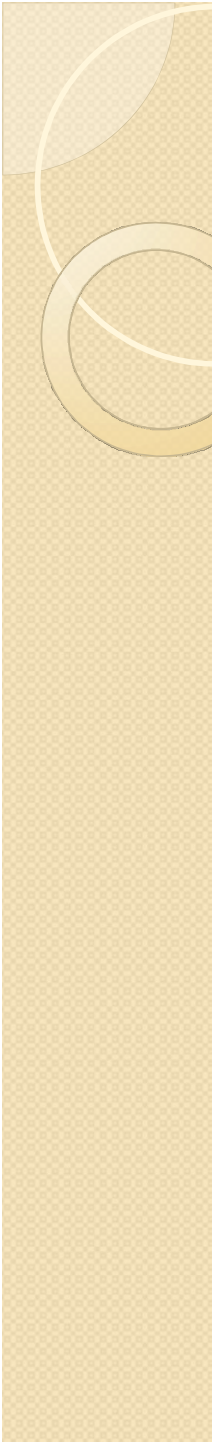
Applications:

Google Translate

International business

Travel

Education



Activity: Translation Challenge

Enter a English sentence.

Translate using:

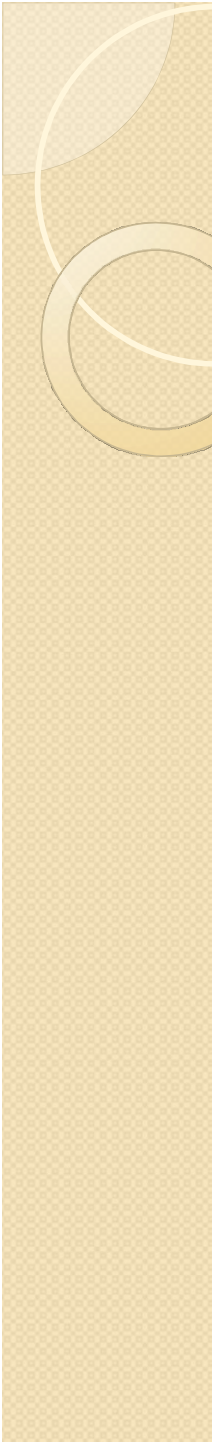
Google Translate

ChatGPT

Then translate back to English.

Discuss:

Did the meaning stay the same?



Hugging Face

Definition

A platform that hosts thousands of AI models, datasets, and interactive demos.

Sections:

Models

Datasets

Spaces

Explain:

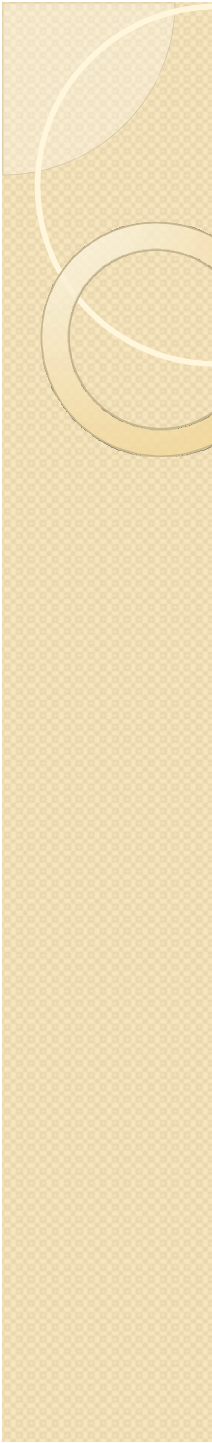
"Think of Hugging Face as GitHub for AI."

Practical Exploration

Divide the class into six groups.

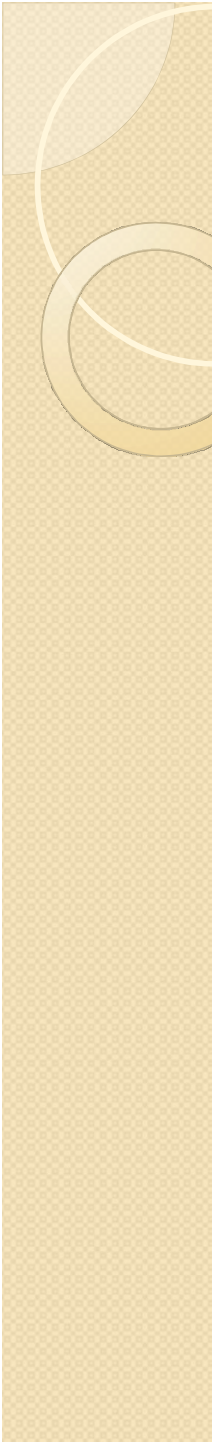
S.No.	Task
1	Sentiment Analysis
2	Translation
3	Named Entity Recognition
4	Text Summarization
5	Question Answering
6	Chatbot

- Open a Hugging Face Space.
- Test five examples.
- Identify one limitation.



Industry Applications

Technology	Company Example
Sentiment Analysis	Amazon Reviews
Translation	Google Translate
NER	News & Banking
Chatbots	OpenAI, Microsoft
OCR	Google Lens
Summarization	Notion AI



Challenge

Designs one AI idea using NLP.

Examples:

AI Resume Analyzer

AI Grammar Checker

AI Email Generator

AI News Summarizer

AI Medical Chatbot

AI Interview Assistant

What problem does it solve?

Which NLP tasks are involved?

Which AI tools would they use?